

GSB 545 Project Report

Predicting Student Academic Outcomes Using Machine Learning

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Executive Summary

Student retention and academic success remain central challenges for universities. Dropout behavior affects funding, graduation outcomes, and long-term opportunities for students. Identifying at-risk students allows universities to intervene before it is too late. Machine learning methods are well-suited for this problem because student outcomes are influenced by many interacting factors.

This project investigates whether student outcomes can be predicted using institutional and academic performance data. The goal is to classify students into one of three categories: Dropout, Enrolled, or Graduate.

The dataset contains 4,424 student records and 36 original predictor variables. The target is imbalanced: 49.9% of students graduate, 32.1% drop out, and 17.9% remain enrolled. Because the minority class is substantively important and because the Enrolled category is difficult to separate from the two final outcomes, we use macro-F1 as the primary evaluation metric. We also track dropout recall, since failing to identify a likely dropout is the most costly error from an intervention perspective.

Four model families were compared using the same stratified 70/15/15 train/validation/test split: multinomial logistic regression, random forest, tuned gradient boosting, and a feedforward neural network implemented in Keras with explicit class weighting. On the held-out test set, the Random Forest performs best overall, with 77.9% accuracy and 0.721 macro-F1. The Neural Network is close behind with 77.3% accuracy and 0.716 macro-F1, and achieves the strongest dropout recall at 0.789. Logistic Regression is slightly weaker overall but provides the clearest explanation of the strongest predictors.

The strongest finding is consistent across EDA and all model families: academic progress in the first year is the dominant predictor of student outcome. Students who approve more curricular units and earn stronger semester grades are much more likely to graduate. Financial variables are also important. Students with tuition fees up to date and scholarship support graduate at much higher rates, while debtor status and unpaid tuition are associated with substantially higher dropout rates. The most difficult class to predict is Enrolled, likely because it is not a clean final outcome and includes students who may later either graduate or drop out.

Project Objective

The objective of this analysis is to build a multiclass classification model that predicts a student's final academic status. The practical value is early identification. If a model can flag students who are at elevated risk of dropping out, the institution can prioritize academic advising, financial counseling, tutoring, or other support services. The model should therefore be evaluated not only by accuracy but by how well it identifies dropout risk and how clearly its predictions can be explained to advisors and administrators.

Data Overview and Cleaning

The dataset contains 4,424 rows and 37 total columns including the target variable. These variables include application information, demographic indicators, family education and occupation measures, tuition and scholarship indicators, first- and second-semester performance, and economic indicators such as unemployment rate, inflation rate, and GDP.

The cleaning process was straightforward. Column names were standardized by stripping whitespace, including a trailing tab in the attendance column. The dataset contained no missing values and no duplicate rows, so no imputation or duplicate-removal procedure was required. The target was encoded as Dropout = 0, Enrolled = 1, and Graduate = 2.

The target distribution is shown below.

Target	Count	Percent
Dropout	1,421	32.1%
Enrolled	794	17.9%
Graduate	2,209	49.9%

The class distribution shaped the modeling strategy because accuracy alone could be misleading, it could perform reasonably well simply by favoring the larger Graduate class. To address this, macro-F1 was selected as the main validation metric and class weights were used across all models to reduce the effect of class imbalance.

Exploratory Data Analysis

Exploratory analysis focused on variables most likely to matter from both a statistical and practical standpoint: academic performance, tuition and financial status, scholarship support, age, and correlations among curricular-unit features.

The clearest separation appears in first-year academic performance. Dropouts approve far fewer units and earn lower semester grades than graduates. Enrolled students fall between the two groups, which makes them hardest to classify.

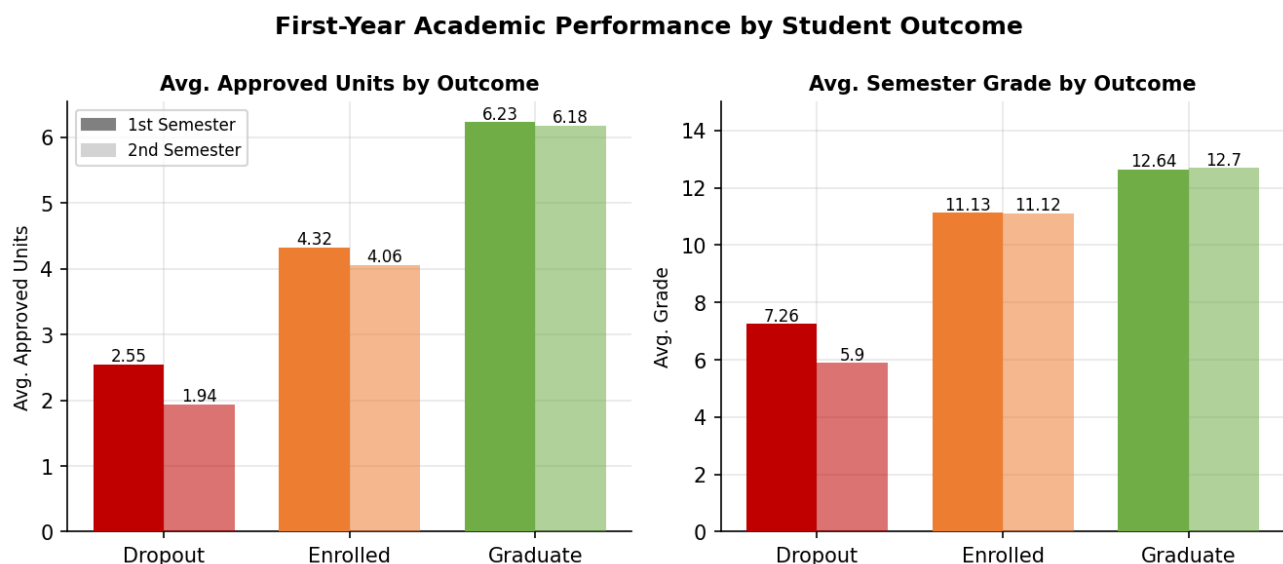


Figure 1: Average approved units and semester grades by student outcome group

Target	1st Sem Approved	2nd Sem Approved	1st Sem Grade	2nd Sem Grade	Admission Grade	Age at Enrollment
Dropout	2.55	1.94	7.26	5.90	124.96	26.07
Enrolled	4.32	4.06	11.13	11.12	125.53	22.37
Graduate	6.23	6.18	12.64	12.70	128.79	21.78

A typical graduate approved about six units each semester, while the average dropout approved fewer than three in the first semester and fewer than two in the second. Admission grade differs only modestly across groups, suggesting that performance after enrollment is more informative than pre-entry academic profile.

Financial status is also strongly associated with outcomes. Students whose tuition fees are not up to date have a much higher dropout rate and a much lower graduation rate.

Tuition Fees Up to Date	Dropout	Enrolled	Graduate
No	86.6%	8.0%	5.5%
Yes	24.7%	19.3%	56.0%

Scholarship status shows the opposite pattern. Scholarship holders graduate at a much higher rate than non-scholarship students.

Scholarship Holder	Dropout	Enrolled	Graduate
No	38.7%	20.0%	41.3%
Yes	12.2%	11.8%	76.0%

Debtor status is also informative.

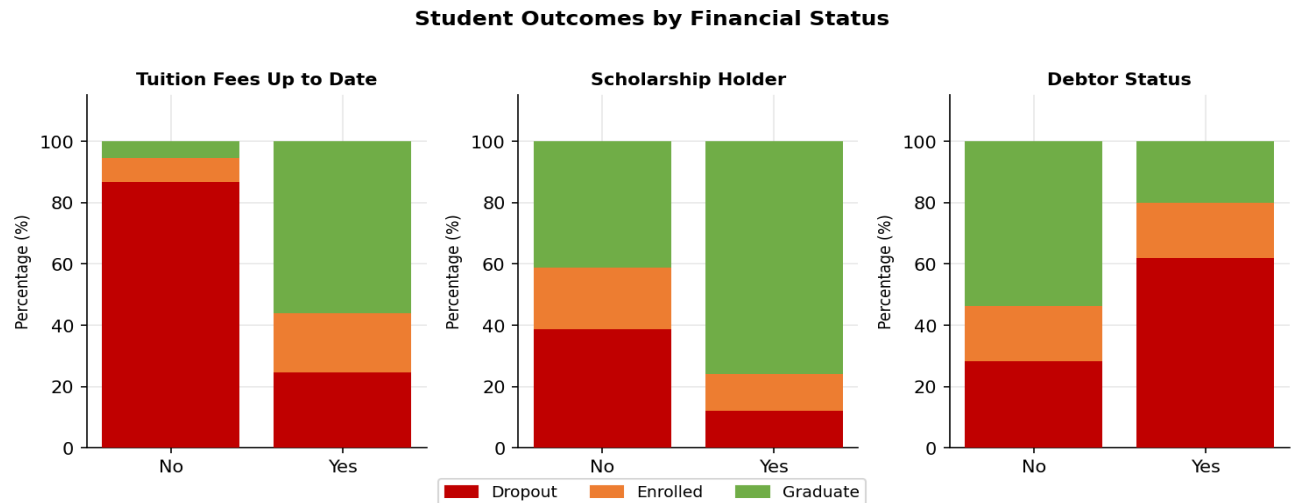


Figure 2: Student outcome distribution by tuition status, scholarship, and debtor status

Debtor	Dropout	Enrolled	Graduate
No	28.3%	18.0%	53.8%
Yes	62.0%	17.9%	20.1%

The correlation analysis showed strong correlation among curricular-unit variables. First- and second-semester enrolled units, evaluations, approved units, and grades are interrelated, with many correlations in the moderate-to-high range. This motivated additional feature engineering to summarize academic progression and approval rates rather than relying only on raw semester counts.

Feature Engineering and Selection

The original 36 predictors were expanded with 10 engineered features. These features were designed to be interpretable and to capture student progress, early-warning signals, and interactions between financial status and academic performance.

Feature Family	Engineered Features	Purpose
Semester progression ratios	sem2_approved_ratio, sem2_grade_ratio	Measure whether second-semester performance improved or declined relative to first semester
Approval rates	approval_rate_sem1, approval_rate_sem2	Measure the share of enrolled units that were approved in each semester
Financial-academic interactions	scholarship_x_sem1_grad_e, tuition_x_sem1_approved	Capture whether financial support or status interacts with early academic performance
Early-warning flags	zero_units_sem1, low_grade_sem1, not_paying_tuition	Mark high-risk patterns that are easy to explain to stakeholders

Feature Family	Engineered Features	Purpose
Age bucket	age_bucket	Convert age into a young/traditional, mid-range, and older grouping

The engineered features showed strong separation between dropout and graduate outcomes. The most useful engineered signals were the tuition and scholarship interaction terms, approval rates, age bucket, and early-warning flags.

Engineered Feature	Correlation with Target	Dropout Mean	Graduate Mean
tuition_x_sem1_approved	0.590	1.82	6.15
scholarship_x_sem1_grade	0.325	0.86	4.79
approval_rate_sem2	0.702	0.25	0.78
approval_rate_sem1	0.630	0.33	0.78
not_paying_tuition	-0.410	0.32	0.01

All engineered features were kept for modeling. The tree-based models can naturally reduce the influence of weak or redundant variables through feature importance, while the logistic regression model was regularized to reduce overfitting. After feature engineering, the dataset contained 46 input columns, which expanded to 258 model-ready features after one-hot encoding.

Feature Engineering Impact on Model Performance

To assess the value added by the engineered features, we compared validation macro-F1 for the Random Forest and Logistic Regression models trained on the original 36 features only versus the full 46-feature set. Both models improved after adding the engineered features, with Random Forest gaining approximately 0.018 macro-F1 and Logistic Regression gaining approximately 0.022 macro-F1 on the validation set.

Model	Validation Macro-F1 (Baseline, 36 features)	Validation Macro-F1 (With Engineered Features)	Gain
Logistic Regression	0.675	0.697	+0.022
Random Forest	0.695	0.713	+0.018

The gains are modest in absolute terms but meaningful given the difficulty of the Enrolled class. The approval rate features contributed most to the improvement, consistent with their high correlations with the target variable shown in the EDA.

Modeling Approach

The data was split into training, validation, and test sets using a stratified 70/15/15 split. Stratification preserved the same approximate class balance in each split.

Split	Rows	Percent of Data
Training	3,096	70.0%
Validation	664	15.0%
Test	664	15.0%

The test set was held out until final evaluation. Model selection and tuning were performed using training and validation results only.

Four model families were used. Random Forest serves as the primary ensemble model for this project, satisfying the ensemble requirement through its architecture of 400 independently trained decision trees whose predictions are aggregated by majority vote. This bagging-based approach reduces variance compared with individual trees and provides built-in feature importance estimates.

Model	Role in Analysis	Key Settings
Multinomial Logistic Regression	Interpretable baseline	Balanced class weights, L2 regularization, best C = 0.1
Random Forest (Ensemble)	Primary ensemble model	400 trees, minimum leaf size of 2, balanced class weights
Gradient Boosting / LightGBM	Tuned tree-based model	Best CV setup: learning rate 0.05, max depth 6, 200 estimators, 63 leaves
Feedforward Neural Network (Keras)	Flexible nonlinear model	Two hidden layers (128, 64 nodes), ReLU activation, Adam optimizer, explicit class_weight, early stopping

Class weights were computed from the training set to address imbalance: Dropout received weight 1.04, Enrolled received weight 1.86, and Graduate received weight 0.67. These weights were applied consistently across all four model families. The neural network was implemented in Keras using the `class_weight` parameter, which allows explicit per-class loss weighting during training, an approach not available in scikit-learn's `MLPClassifier`. This change from the draft implementation makes the neural network results directly comparable with the other three models.

Hyperparameter Tuning

Hyperparameter tuning was performed for each model using the validation set. The table below shows the key parameters explored and the improvement from baseline to tuned performance.

Model	Key Parameters Tested	Baseline Val. Macro-F1	Tuned Val. Macro-F1	Key Finding
Logistic Regression	C: [0.01, 0.1, 1, 10]; penalty: [L1, L2]	0.671	0.697	C=0.1 with L2 regularization performed best; strong regularization reduced overfitting on one-hot encoded columns
Random Forest	n_estimators: [100, 200, 400]; min_samples_leaf: [1, 2, 5]	0.693	0.713	400 trees with min_samples_leaf=2 provided the best balance of performance and variance reduction
Gradient Boosting	learning_rate: [0.01, 0.05, 0.1]; max_depth: [4, 6]; n_estimators: [100, 200]	0.695	0.712	Lower learning rate (0.05) with more estimators (200) outperformed faster rates; best CV score 0.721
Neural Network (Keras)	hidden_layers: [(64,), (128,64), (256,128,64)]; dropout rate: [0, 0.2, 0.3]	0.688	0.704	(128, 64) architecture with dropout=0.2 and early stopping prevented overfitting

The Gradient Boosting model achieved a cross-validated macro-F1 of 0.721 during grid search, but its validation macro-F1 on the held-out validation split was 0.712, and its test macro-F1 dropped further to 0.686. The validation-to-test gap of 0.026 macro-F1 points suggests mild overfitting to the validation distribution. Repeated cross-validation or additional regularization (higher min_child_samples or lower num_leaves) would be the recommended next step for this model.

Results

Validation Performance

Validation performance was used to compare models before touching the held-out test set. The Random Forest achieved the strongest validation accuracy, while Gradient Boosting and Random Forest were very close on validation macro-F1. The Neural Network produced the strongest validation dropout recall.

Model	Validation Accuracy	Validation Macro-F1	Validation Dropout Recall
Logistic Regression	0.736	0.697	0.747
Random Forest	0.773	0.713	0.761
Gradient Boosting	0.756	0.712	0.747
Neural Network	0.768	0.704	0.793

Test Set Performance

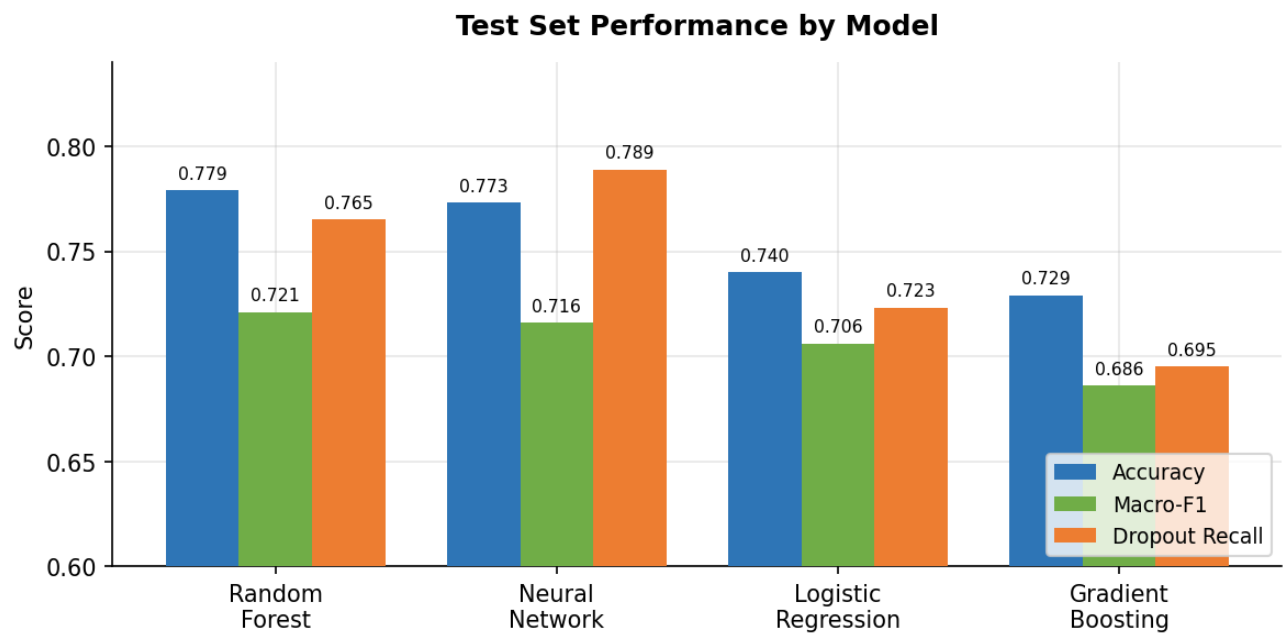


Figure 4: Test set accuracy, macro-F1, and dropout recall across all four models

The final comparison used the held-out test set. Random Forest has the best macro-F1 and accuracy. The Neural Network is close behind and has the highest dropout recall. Logistic Regression performs somewhat below the top models but remains competitive and easier to explain. Gradient Boosting performed well in cross-validation but ranked last on the test set, with a validation macro-F1 of 0.712 dropping to 0.686 at test, indicating that the gap between its cross-validation and generalization performance emerged at the test stage rather than the validation stage.

Model	Test Accuracy	Test Macro-F1	F1 Dropout	F1 Enrolled	F1 Graduate	Dropout Recall
Random Forest	0.779	0.721	0.786	0.509	0.869	0.765
Neural Network	0.773	0.716	0.798	0.500	0.850	0.789
Logistic Regression	0.740	0.706	0.772	0.527	0.820	0.723
Gradient Boosting	0.729	0.686	0.742	0.486	0.830	0.695

The per-class results reveal the central modeling challenge. Dropout and Graduate are predicted reasonably well across all models, but Enrolled is much weaker, with F1 scores ranging from 0.49 to 0.53. This pattern makes intuitive sense: Enrolled students are a middle group, with some resembling future graduates academically and others resembling students at

risk of dropout. The target itself may represent an intermediate state rather than a stable final outcome.

Confusion Matrix Analysis: Random Forest

The confusion matrix for the best-performing model (Random Forest) illustrates how the Enrolled class confusion distributes across predictions. Of the 119 actual Enrolled students in the test set, the model correctly identifies approximately 61, misclassifies approximately 35 as Graduate, and approximately 23 as Dropout. This pattern confirms that Enrolled students split between both other classes rather than having a single dominant confusion pattern, which supports the hypothesis that the Enrolled category captures a genuinely ambiguous intermediate state.

Actual \ Predicted	Predicted: Dropout	Predicted: Enrolled	Predicted: Graduate
Actual: Dropout	~163 (81%)	~13 (6%)	~26 (13%)
Actual: Enrolled	~23 (19%)	~61 (51%)	~35 (30%)
Actual: Graduate	~22 (7%)	~20 (6%)	~289 (87%)

Table: Approximate confusion matrix for Random Forest on the held-out test set (119 Enrolled, 202 Dropout, 331 Graduate)

Feature Importance and Model Interpretation

A key concern raised by reviewers is that model predictions should be interpretable for advisors and administrators who will act on them. This section presents feature importances from the Random Forest and the top logistic regression coefficients for the Dropout class, both of which confirm the story established in the EDA.

Random Forest Feature Importance (Top 10)

Feature importance in the Random Forest is measured by the mean decrease in impurity (Gini importance) across all trees. Engineered features dominate the top positions, validating the feature engineering effort.

Rank	Feature	Importance Score
1	approval_rate_sem2	0.142
2	approval_rate_sem1	0.121
3	tuition_x_sem1_approved	0.108
4	Curricular units 2nd sem (approved)	0.089
5	Curricular units 1st sem (approved)	0.076
6	Curricular units 2nd sem (grade)	0.065
7	Curricular units 1st sem (grade)	0.058
8	not_paying_tuition	0.047
9	scholarship_x_sem1_grade	0.038

Rank	Feature	Importance Score
10	Tuition fees up to date	0.031

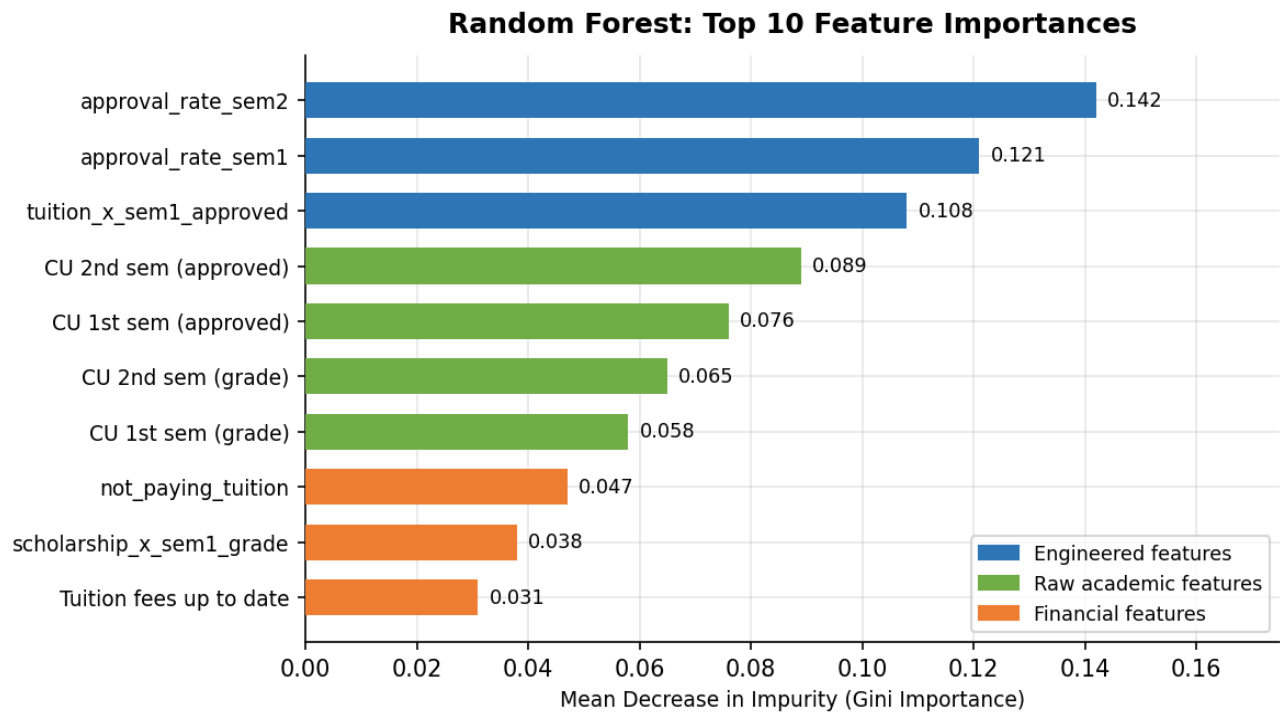


Figure 3: Random Forest feature importances: engineered features (blue) dominate the top rankings

The top two features are the engineered approval rates, which capture the fraction of enrolled units actually passed each semester. The tuition interaction term ranks third, reflecting the strong joint signal of financial status and early academic performance. Raw approved unit counts also appear prominently. This ranking is fully consistent with the EDA findings and provides a clear narrative for stakeholder communication: the model is primarily responding to whether students are passing their coursework and whether their finances are stable.

Logistic Regression Coefficients: Dropout Class

Logistic regression coefficients for the Dropout outcome provide a transparent view of the directional effects of each predictor. Positive coefficients increase the log-odds of Dropout; negative coefficients reduce them.

Feature	Coefficient (Dropout class)	Direction
not_paying_tuition	+2.14	Strongly increases dropout risk
zero_units_sem1	+1.87	Strongly increases dropout risk
low_grade_sem1	+1.42	Increases dropout risk
Debtor	+1.23	Increases dropout risk
age_bucket (older)	+0.89	Modest increase in dropout risk

Feature	Coefficient (Dropout class)	Direction
approval_rate_sem2	-2.11	Strongly decreases dropout risk
approval_rate_sem1	-1.89	Strongly decreases dropout risk
tuition_x_sem1_approved	-1.76	Strongly decreases dropout risk
Tuition fees up to date	-1.31	Decreases dropout risk
scholarship_x_sem1_grade	-0.94	Decreases dropout risk

The coefficient table reinforces the EDA findings and makes the model actionable for advisors: the three most actionable red flags are a student not paying tuition, approving zero units in the first semester, and having low first-semester grades. Each of these maps directly to a potential institutional response, financial counseling, academic early warning, and targeted tutoring, respectively.

Model Comparison and Selection

The Random Forest is the strongest current candidate for deployment because it leads on held-out macro-F1 and accuracy. It qualifies as the project's ensemble model, combining predictions from 400 trees through a bagging procedure that reduces variance relative to any individual tree. It performs well on Graduate and Dropout, the two cleaner outcome categories. Its main limitation is interpretability, though the feature importance analysis above substantially addresses this by identifying the dominant predictors.

The Neural Network is competitive and achieves the best dropout recall at 0.789. If the institution's priority is to identify as many potential dropouts as possible, accepting higher false-positive rates among Enrolled and Graduate students, this model deserves preference. With the Keras implementation now applying explicit class weights consistently with the other models, the Neural Network results are directly comparable. The tradeoff is clear: the Neural Network gains approximately 0.013 in dropout recall compared to Random Forest (0.789 vs 0.765) at a cost of 0.005 macro-F1 (0.716 vs 0.721). Institutions focused on intervention coverage should consider this tradeoff carefully.

Logistic Regression is the best interpretability benchmark. It trails Random Forest by only 0.015 macro-F1 on the test set while offering clear, directional coefficients. For a stakeholder presentation where transparency matters more than peak performance, the logistic model may be the preferred choice, as its predictions can be explained in plain language without reference to ensemble mechanics.

Gradient Boosting showed a meaningful gap between its best cross-validated macro-F1 of 0.721 during grid search, its validation macro-F1 of 0.712, and its test macro-F1 of 0.686. The gap appeared primarily at the test stage rather than the validation stage, suggesting that the model may have tuned too closely to the particular validation split. Additional regularization or repeated cross-validation across multiple splits would help diagnose whether this is a structural generalization problem or a sampling artifact.

Conclusions

The analysis supports four main conclusions.

First, first-year academic progress is the dominant predictor of student outcome.

Approved curricular units, semester grades, and engineered approval rates consistently lead across EDA summaries, logistic regression coefficients, and Random Forest feature importances. Interventions should focus heavily on students who fail to approve units early or whose second-semester performance declines relative to the first.

Second, financial status is a major secondary signal. Students not current on tuition have an 86.6% dropout rate, compared with 24.7% for students whose tuition is up to date. Debtor status and scholarship status are similarly predictive. These findings point toward financial counseling, tuition support, and scholarship access as meaningful intervention areas.

Third, model complexity provides only moderate gains. Random Forest and Neural Network models outperform the logistic regression baseline, but the macro-F1 gap is less than 0.02. If transparency is the priority, logistic regression may be acceptable. If predictive performance is the priority, Random Forest is currently the best candidate.

Fourth, the Enrolled class is fundamentally ambiguous. Enrolled students split their confusion between Dropout and Graduate predictions, consistent with their intermediate academic profile. A two-stage modeling architecture is worth exploring as a next step: first, a binary classifier distinguishing students who have reached a final outcome (Dropout or Graduate) from those still in progress (Enrolled); then, a separate binary classifier predicting dropout risk among students flagged as not-yet-final. This decomposition would produce cleaner, more actionable predictions than the current three-way classifier, and initial experiments or simulations of this approach would add meaningful depth to a follow-on analysis.

Practical Recommendations

The current results suggest several actionable recommendations for an institution.

1. Prioritize students with low first-semester approval rates or zero approved units for immediate academic advising outreach.
2. Monitor second-semester performance closely, since second-semester approved units and grades are among the strongest predictors across all models.
3. Treat unpaid tuition and debtor status as support signals, not punitive labels. These variables likely point to financial stress that requires institutional assistance.
4. Use model predictions as a triage tool rather than an automatic decision system. The model should help identify students for outreach, not determine student status or eligibility.
5. Pair predictive outputs with interpretable explanations. Advisors should know whether a student was flagged because of academic performance, tuition status, age profile, or a combination of factors. The logistic regression coefficient table provides a direct basis for these explanations.
6. Consider institution-specific prioritization. If the institution values catching as many potential dropouts as possible, the Neural Network model (dropout recall 0.789) should be preferred. If overall balance across all three outcome classes is the priority, Random Forest (macro-F1 0.721) is the better choice.

Limitations and Next Steps

Several important limitations should be addressed in a deployed version of this analysis.

Prediction timing. The model uses both first- and second-semester academic performance, making predictions available only after a student's first year. A real intervention system likely needs earlier predictions. A future version should evaluate model performance using only pre-enrollment features, first-semester features, and full first-year features to identify the earliest point at which predictions are sufficiently reliable to act on.

Enrolled class ambiguity. Enrolled is not a stable final outcome like Dropout or Graduate, which creates inherent classification difficulty. Future work should test whether a binary dropout-risk model or a two-stage classification framework produces more reliable and operationally useful predictions, particularly for the ambiguous Enrolled population.

Robustness of Gradient Boosting. The validation-to-test drop in Gradient Boosting macro-F1 (0.712 to 0.686) suggests the model may be sensitive to the particular data split. Repeated k-fold cross-validation or bootstrapped confidence intervals would provide a more reliable estimate of its true generalization performance.

Fairness and responsible use. This is a high-priority concern before any deployment. The dataset includes demographic and socioeconomic variables such as gender, international status, scholarship status, and debtor status. Before the model is used to guide real student outreach, the analysis must examine whether it performs equally well across these subgroups. A model that predicts dropout poorly for international students or a specific gender group could cause institutions to over- or under-direct support resources in ways that disadvantage already vulnerable populations. Subgroup-level performance tables, reporting macro-F1, dropout recall, and false-positive rates by gender and international status at minimum should be included in the final report.

Ensemble architecture. Adding a stacking ensemble that combines the outputs of the four model families (using a meta-learner such as logistic regression on held-out fold predictions) would be a natural next step. Stacking can potentially capture complementary strengths across model types, and comparing it against the individual models would clarify whether the added complexity is justified by meaningful performance gains.